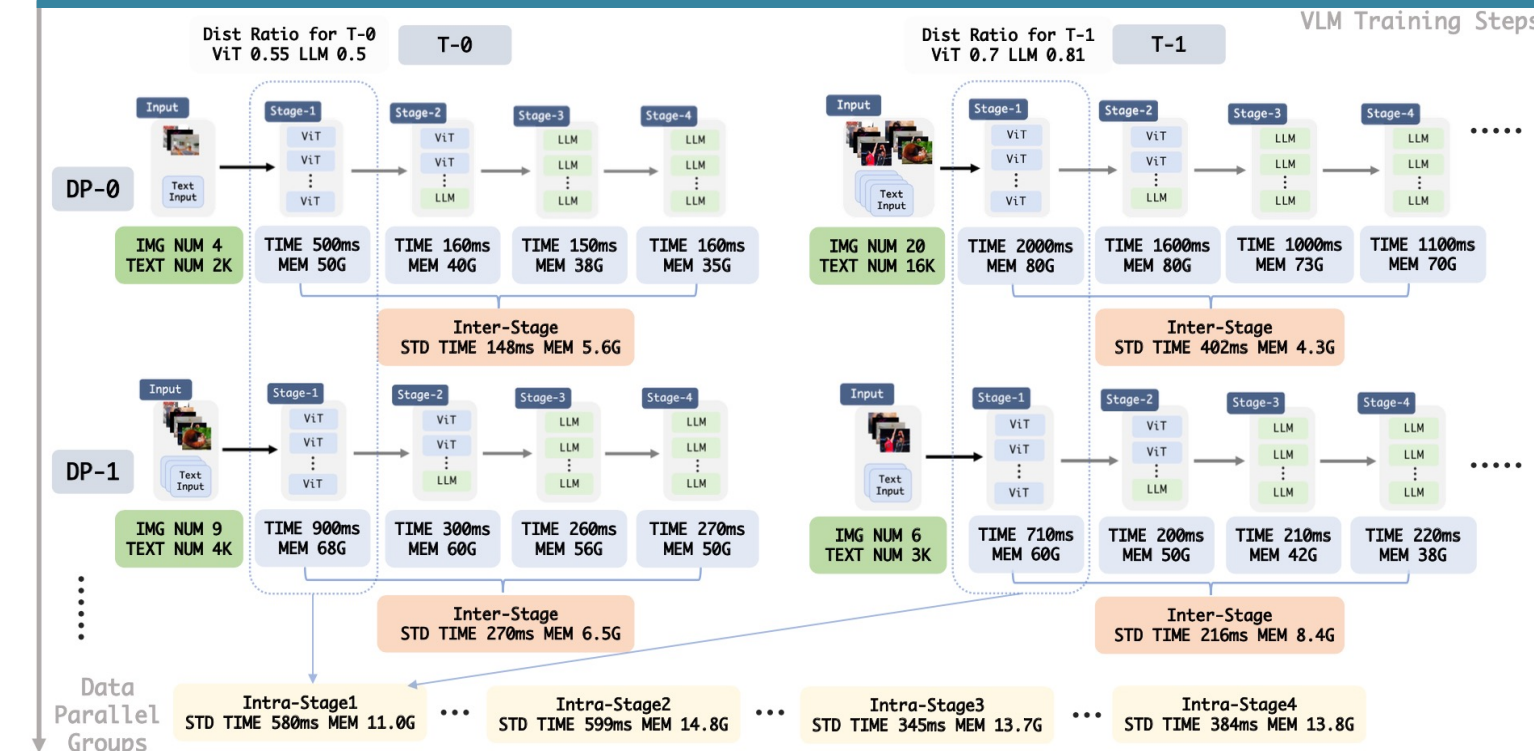


OmniBal: Towards Fast Instruction-Tuning for Vision-Language Models via Omniverse Computation Balance

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Problems

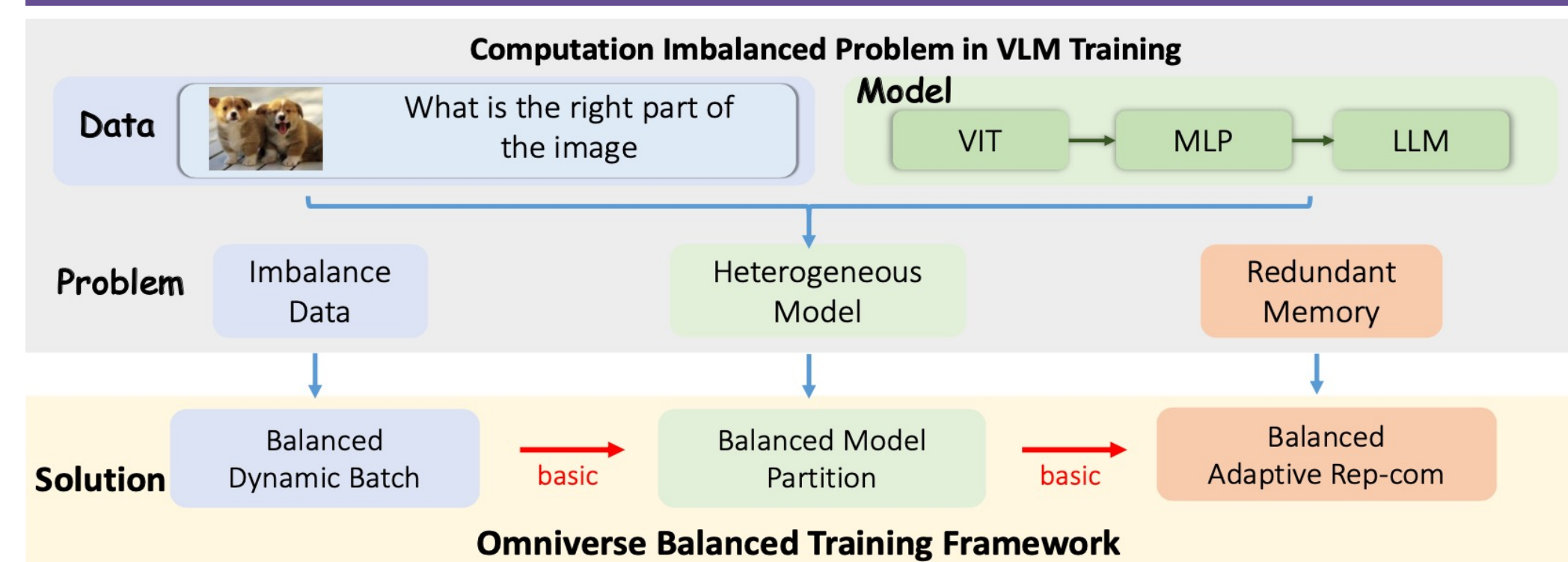


- **Inter-Stage:** computation imbalance of different pipeline parallel stages.
- **Intra-Stage:** indicates the computation imbalance of the same stage across time and devices.

Motivations & Challenges

- **Data Imbalance:** VLMs handle texts and images requiring data integrity, leading to varying input sizes.
- **Model Imbalance:** Structural differences between VIT and LLM result in different computational demands.
- **Memory Imbalance:** VLMs encounter memory challenges due to variable data inputs and model heterogeneity.

Method: framework



DATA

Algorithm 1 ISF: Iterative Sampling and Filtering

```

1: ISF: Sampling Stage
2:  $\mathcal{D} = \text{randperm}(\mathcal{D})$ , set  $\mathcal{G} = []$ 
3: for  $(x_i, y_i)$  in  $\mathcal{D}$  do
4:    $\mathcal{G} \leftarrow \mathcal{G} + (x_i, y_i)$ 
5:   if  $I_v > Q_v$  or  $I_t > Q_t$  then
6:      $\mathcal{G} \leftarrow \mathcal{G} - (x_i, y_i)$ 
7:      $\mathcal{P} \leftarrow \mathcal{P} + \mathcal{G}$ , set  $\mathcal{G} = [(x_i, y_i), ]$ 
8:   end if
9: end for
10: return:  $\mathcal{P}$ 

```

```

11: ISF: Filtering Stage
12: Get  $\mathcal{P}$  from Sampling Stage
13: for  $\mathcal{G}$  in  $\mathcal{P}$  do
14:   if  $I_v < Q'_v$  and  $I_t < Q'_t$  then
15:      $\mathcal{P} \leftarrow \mathcal{P} - \mathcal{G}$ 
16:   else
17:     remove all  $(x_i, y_i)$  of  $\mathcal{G}$  from  $\mathcal{D}$ 
18:   end if
19: end for
20: return  $\mathcal{P}, \mathcal{D}$ 

```

MODEL

- Profiling VLM layer time and p2p communication
- Generate base anchor partitions
- Generate candidate partitions around anchors
- Search the best partition based on layer time and p2p-communication

MEMORY

- Profiling VLM layer memory cost for Vision and LLM input
- Estimate memory cost for VLM layer without re-computation
- Get the optimal re-computation strategy based on memory cost

Experiments

➤ Main Results

Model	Balance?	Backend	MMB-EN/CN	ChartQA	AI2D	MMVet	MME	GPU Days (speed-up)
6+20B	×	DeepSpeed	78.2/77.4	86.2	71.3	48.9	1901.2	38.9 (1.00×
	✓	DeepSpeed	78.7/77.6	86.5	71.4	50	1969.4	25.3 (1.54×)
	×	Megatron	79.5/77.7	87.3	71.6	45.0	1957.7	61.8 (0.63×
	✓	Megatron	78.6/77.5	86.7	70.9	48.5	1956.3	21.3 (1.83×)
6+34B	×	DeepSpeed	80.0/79.2	86.6	73.4	45.9	2015.8	54.3 (1.00×
	✓	DeepSpeed	80.9/79.0	89.1	73.3	47.0	2153.6	35.5 (1.53×)
	×	Megatron	80.2/79.3	88.9	73.7	44.2	2111.9	75.4 (0.72×
	✓	Megatron	80.1/78.0	89.3	73.5	45.4	2072.7	30.5 (1.80×)

➤ Data Results

Method	AVE-BS	Max-Seq-Len		Pad Ratio	Dist Ratio		Balanced	GPU Days	
		VIT	LLM		VIT	LLM		w/o MB	w MB
baseline	4	20K	16K	0.31	0.34	0.30	×	61.8	42.2
length-group	4	20K	16K	0.20	0.26	0.13	×	54.0	40.0
device-group	4	20K	16K	0.378	0.125	0.15	×	54.5	43.6
ISF(ours)	4.6	9K	4K	0	0.02	0.14	✓	51.9	29.0

➤ Model Results

Method	VAR(param)	VAR(num.layer)	VAR(fwd.time)	Δ SUM(comm)	GPU days
(1) parameter-based	0.03	13.4	93.6	+0.0	42.2
(2) layer-based	0.64	1.2	20.1	+8.2	30.6
(3) profile-based	0.85	2.1	6.5	+16.6	30.9
(4) BMP (ours)	0.83	<u>1.5</u>	<u>12.2</u>	-21.0	29.0

➤ Memory Results

Method	V-Seq-Len	L-Seq-Len	VRAM ₁	VRAM ₂	VRAM ₃	VRAM ₄	GPU Days
baseline	4K~20K	1K~16K	13~50.2	7.3~40.5	7.3~40.5	7.3~40.5	61.8
+ data & model balance	9K	4K	58.2	56.2	32.5	32.7	29.0
+ memory balance	9K	4K	12.3	21.7	24.7	30.0	21.3

Conclusion

- **Balances 3D parallel workloads** to eliminate computational imbalance in large-scale VLM training.
- **1.8 × end-to-end speed-up** demonstrated on open-source models.
- **Model-, dataset- & hardware-agnostic**, making it easy to deploy across diverse setups.

Code

<https://github.com/ModelTC/OmniBal>